



Original research



Open-label, phase II study of talimogene laherparepvec plus pembrolizumab for the treatment of advanced melanoma that progressed on prior anti-PD-1 therapy: MASTERKEY-115,^{1,2}

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ABSTRACT

Background: Treatment options for immunotherapy-refractory melanoma are an unmet need. The MASTERKEY-115 phase II, open-label, multicenter trial evaluated talimogene laherparepvec (T-VEC) plus pembrolizumab in advanced melanoma that progressed on prior programmed cell death protein-1 (PD-1) inhibitors.

Methods: Cohorts 1 and 2 comprised patients (unresectable/metastatic melanoma) who had primary or acquired resistance, respectively, and disease progression within 12 weeks of their last anti-PD-1 dose. Cohorts 3 and 4 comprised patients (resectable disease) who underwent complete surgery, received adjuvant anti-PD-1, and experienced recurrence. Cohort 3 were disease-free for < 6 months and cohort 4 for ≥ 6 months after starting the adjuvant anti-PD-1 therapy and before confirmed recurrence. The primary endpoint was objective response rate (ORR) per RECIST v1.1. Secondary endpoints included complete response rate (CRR), disease control rate (DCR) and progression-free survival (PFS) per RECIST v1.1 and irRC-RECIST, and safety.

Results: Of the 72 enrolled patients, 71 were treated. The ORR (95% CI) was 0%, 6.7% (0.2–32.0), 40.0% (16.3–67.7), and 46.7% (21.3–73.4) in cohorts 1–4, respectively; iORR was 3.8% (0.1–19.6), 6.7% (0.2–32.0), 53.3% (26.6–78.7), and 46.7% (21.3–73.4). iCRR was 0%, 0%, 13.3%, and 13.3%. Median iPFS (months) was 5.5, 8.2, not estimable [NE], and NE for cohorts 1–4, respectively; iDCR was 50.0%, 40.0%, 73.3%, and 86.7%. Treatment-related adverse events (TRAEs), grade ≥ 3 TRAEs, serious AEs, and fatal AEs occurred in 54 (76.1%), 9 (12.7%), 24 (33.8%), and 10 (14.1%) patients, respectively.

List of Abbreviations: AE, adverse event; BOR, best overall response; CI, confidence interval; CR, complete response; CRR, complete response rate; CTCAE, Common Terminology Criteria for Adverse Events; CTLA-4, cytotoxic T-cell lymphocyte-4; DCR, disease control rate; DOR, duration of response; ECOG, Eastern Cooperative Oncology Group; GM-CSF, granulocyte-macrophage colony-stimulating factor; HR, hazard ratio; ICI, immune checkpoint inhibitor; irAE, immune-related adverse event; irRC-RECIST, immune-related RECIST; ITT, intent-to-treat; LDH, lactate dehydrogenase; NE, not estimable; OR, Odds ratio; ORR, objective response rate; OS, overall survival; PD-1, programmed cell death-1; PD, progressive disease; PFU, plaque-forming unit; PFS, progression-free survival; PR, partial response; Q3W, every 3 weeks; RECIST, Response Evaluation Criteria in Solid Tumors; SAE, serious adverse event; SD, stable disease; SOC, standard of care; TEAE, treatment-emergent adverse event; TIL, tumor-infiltrating lymphocyte; TRAE, treatment-related adverse event; T-VEC, talimogene laherparepvec.

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Conclusion: T-VEC-pembrolizumab demonstrated antitumor activity and tolerability in PD-1–refractory melanoma, specifically in patients with disease recurrence on or after adjuvant anti-PD-1.

Trial registration: ClinicalTrials.gov identifier - NCT04068181

1. Introduction

Immune checkpoint inhibition (ICI) with programmed cell death protein-1 (PD-1) inhibitors (pembrolizumab and nivolumab) and combination anti-PD-1/cytotoxic T-cell lymphocyte-4 (CTLA-4) (nivolumab plus ipilimumab) or nivolumab plus anti-LAG-3 (nivolumab-relatlimab) are the standard of care (SOC) for advanced melanoma in the frontline metastatic setting [1–4]. Durable survival and response rates are clinical hallmarks of ICI therapy, yet > 50% of patients fail to ever respond (primary resistance) to anti-PD-1 monotherapy, and approximately 40% of those who initially respond experience disease recurrence (acquired resistance) within 3 years [5–8].

For high-risk resected stage III/IV melanoma, adjuvant anti-PD-1 is considered a SOC [9–11]. Yet, as in the metastatic frontline setting, these patients may experience disease recurrence [5]. The proportion of patients with metastatic melanoma who received adjuvant anti-PD-1 will increase, generating an emerging unmet need [5]. Yet, there is a paucity of trials that have addressed treatment for this population.

Combination ICI is an effective treatment for unresectable or metastatic melanoma following disease progression on anti-PD-1 or anti-PDL-1, with the combinations of ipilimumab with nivolumab or pembrolizumab showing improved efficacy versus ipilimumab [12–14]. Newer approaches have shown promising responses in PD-1/L1–refractory advanced melanoma; an objective response rate (ORR) of 21.4% was shown for lenvatinib plus pembrolizumab; an ORR of 31.4% reported for lifileucel, an autologous tumor-infiltrating lymphocyte (TIL) cell therapy; and nivolumab plus relatlimab demonstrated durable clinical activity in heavily pretreated advanced melanoma, with ORRs ranging from 9.2% to 12.0% across cohorts [15–17]. Lenvatinib and lifileucel are associated with toxicities and uncertain durability, and lifileucel was limited to select populations for clinical rescue [15,16].

Combination immunotherapy induces long-lasting toxicities [18–20]. Effective combination strategies that enhance efficacy without additional toxicity are thus highly needed.

Talimogene laherparepvec (T-VEC) is an oncolytic viral intratumoral immunotherapy that features not only cell-type-unselective viral replication, but also reshaping of cellular composition and tumor microenvironment (TME), shifting the tide towards immune activation as demonstrated in samples from patients treated with T-VEC. Recent phase II/III advanced melanoma trials have shown improved benefits with T-VEC monotherapy (versus GM-CSF) and T-VEC plus ipilimumab (versus ipilimumab) [21–23]. However, T-VEC combined with pembrolizumab did not confer a significant survival benefit versus placebo-pembrolizumab in anti-PD-1 naïve melanoma [24]. Intratumoral therapy is particularly effective for patients with unresectable locoregional melanoma who tend to have lesions amenable to direct injection in a high percentage of cases; locoregional disease with cutaneous, subcutaneous or lymph node involvement develop in up to 10.0% of patients with metastatic melanoma [25–27]. Limited data exist regarding the effectiveness of T-VEC in the ICI-refractory setting [21, 28]. In a retrospective analysis of data from two phase II trials of T-VEC in patients with unresectable metastatic melanoma who had received prior ICI therapy, the ORRs were 23.5% (4/17) and 21.4% (6/28). This analysis, though limited, suggested measurable activity of T-VEC in the post-ICI setting [29].

Combined T-VEC and anti-PD-1 therapy is hypothesized to play complementary roles in regulating adaptive immunity and may overcome ICI failure [30–32]. In a phase Ib study, Ribas et al. demonstrated that patients who responded to combination therapy (T-VEC plus pembrolizumab) had increased CD8+ T cells, elevated programmed death ligand-1 (PD-L1) protein expression, as well as IFN- α gene expression on several tumor cell subsets after T-VEC treatment [30]. Therefore, T-VEC

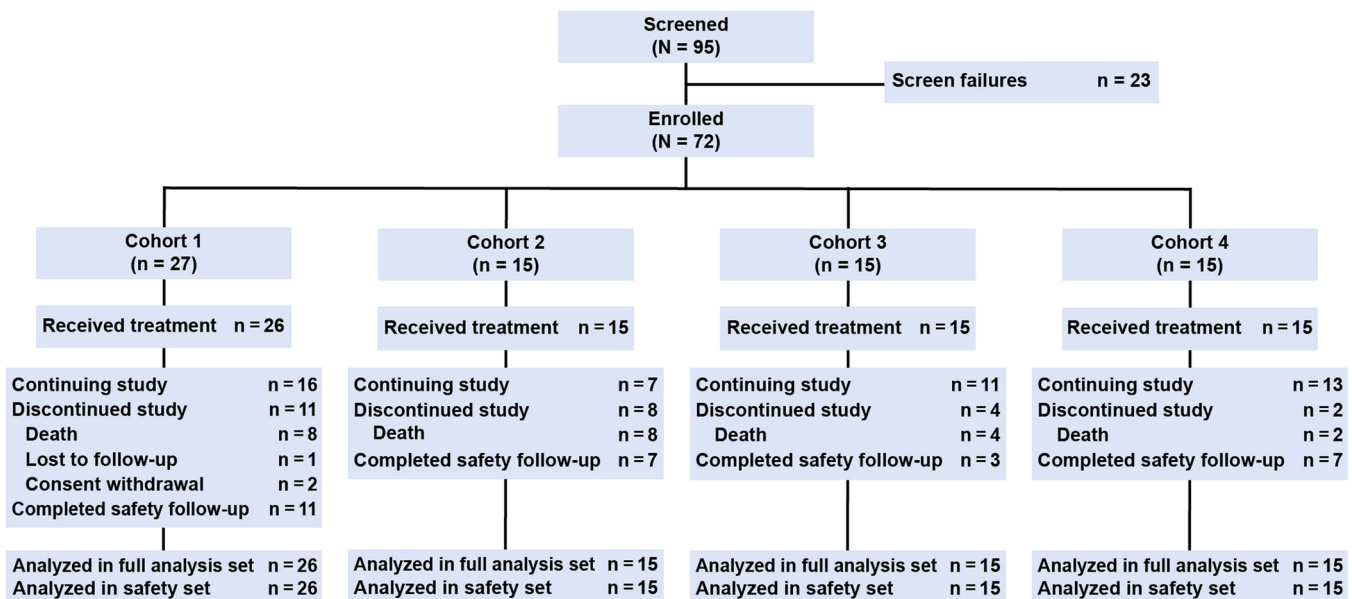


Fig. 1. Patient disposition. Ninety-five patients were screened for the study, and 23 screen failures occurred. Seventy-two patients were enrolled into one of four cohorts to receive T-VEC plus pembrolizumab combination treatment (one patient in cohort 1 was deemed ineligible and was not dosed with any study drug); 71 patients were evaluable for efficacy and safety analyses. Cohorts 1 and 2 comprised patients with primary and acquired resistance, respectively, who received anti-PD-1 in a locally recurrent or metastatic setting and had disease progression within 12 weeks of their last anti-PD-1 dose. Cohorts 3 and 4 comprised patients with resectable disease who received adjuvant anti-PD-1 and were disease-free for < 6 months (cohort 3) or $\geq$ 6 months (cohort 4) before confirmed disease progression. PD-1, programmed cell death protein-1; T-VEC, talimogene laherparepvec.

Table 1
Patient demographics and baseline characteristics.

Characteristic	T-VEC + pembrolizumab				
	Primary resistance Cohort 1 (n = 26)	Acquired resistance Cohort 2 (n = 15)	Adjuvant < 6 months Cohort 3 (n = 15)	Adjuvant ≥ 6 months Cohort 4 (n = 15)	All cohorts (N = 71)
Age, median (range), years	64.5 (34–86)	63.0 (43–89)	58.0 (33–75)	67.0 (46–74)	65.0 (33–89)
Sex, male	18 (69.2)	7 (46.7)	10 (66.7)	13 (86.7)	48 (67.6)
Race					
White	25 (96.2)	14 (93.3)	14 (93.3)	15 (100.0)	68 (95.8)
Other	1 (3.8)	1 (6.7)	1 (6.7)	0	3 (4.2)
ECOG performance status					
0	14 (53.8)	12 (80.0)	15 (100.0)	13 (86.7)	54 (76.1)
1	12 (46.2)	3 (20.0)	0	2 (13.3)	17 (23.9)
Grouped disease stage at baseline					
Stage IIIB–IVM1a	8 (30.8)	7 (46.7)	9 (60.0)	10 (66.7)	34 (47.9)
Stage IVM1b–d	18 (69.2)	8 (53.3)	6 (40.0)	5 (33.3)	37 (52.1)
BRAF ^{V600} mutation status					
Mutation present	11 (42.3)	4 (26.7)	4 (26.7)	1 (6.7)	20 (28.2)
Mutation absent	15 (57.7)	11 (73.3)	11 (73.3)	13 (86.7)	50 (70.4)
Unknown	0	0	0	1 (6.7)	1 (1.4)
Baseline LDH ^a					
≤ 1x ULN	19 (73.1)	7 (46.7)	13 (86.7)	11 (73.3)	50 (70.4)
> 1x ULN	7 (26.9)	8 (53.3)	2 (13.3)	4 (26.7)	21 (29.6)
≤ 2x ULN	26 (100.0)	15 (100.0)	15 (100.0)	14 (93.3)	70 (98.6)
> 2x ULN	0	0	0	1 (6.7)	1 (1.4)
HSV-1 serostatus					
Positive	20 (76.9)	9 (60.0)	10 (66.7)	13 (86.7)	52 (73.2)
Negative	4 (15.4)	5 (33.3)	4 (26.7)	2 (13.3)	15 (21.1)
Missing/Unknown	2 (7.7)	1 (6.7)	1 (6.7)	0 (0.0)	4 (5.6)
Prior PD-1 therapy ^b					
Nivolumab	15 (57.7)	12 (80.0)	14 (93.3)	13 (86.7)	54 (76.1)
Ipilimumab/ Nivolumab	3 (11.5)	3 (20.0)	0	0	6 (8.5)
Pembrolizumab	10 (38.5)	4 (26.7)	1 (6.7)	2 (13.3)	17 (23.9)
Other PD-1 inhibitor	1 (3.8)	1 (6.7)	0	0	2 (2.8)
Prior anticancer therapy	26 (100.0)	15 (100.0)	15 (100.0)	15 (100.0)	71 (100.0)
Number of lines of therapy ^{c,d}					
1	18 (69.2)	5 (33.3)	13 (86.7)	13 (86.7)	49 (69.0)
2	8 (30.8)	5 (33.3)	1 (6.7)	1 (6.7)	15 (21.1)
3	0	5 (33.3)	0	0	5 (7.0)
4	0	0	1 (6.7)	1 (6.7)	2 (2.8)
Number of prior anticancer therapies for	47	48	20	19	134

Table 1 (continued)

Characteristic	T-VEC + pembrolizumab				
	Primary resistance Cohort 1 (n = 26)	Acquired resistance Cohort 2 (n = 15)	Adjuvant < 6 months Cohort 3 (n = 15)	Adjuvant ≥ 6 months Cohort 4 (n = 15)	All cohorts (N = 71)
current malignancy					
Setting for prior anticancer therapies ^e					
Adjuvant	3 (6.4)	4 (8.3)	17 (85.0)	18 (94.7)	42 (31.3)
Metastatic	43 (91.5)	43 (89.6)	3 (15.0)	1 (5.3)	90 (67.2)
Palliative	0	1 (2.1)	0	0	1 (0.7)
Other	1 (2.1)	0	0	0	1 (0.7)
Prior surgery	24 (92.3)	13 (86.7)	15 (100.0)	15 (100.0)	67 (94.4)
Prior radiotherapy	9 (34.6)	1 (6.7)	0	1 (6.7)	11 (15.5)

Data are the number (%) of patients unless specified otherwise. The safety analysis set included all enrolled patients who received ≥ 1 dose of T-VEC or pembrolizumab. ECOG, Eastern Cooperative Oncology Group; HSV-1, herpes simplex virus 1; LDH, lactate dehydrogenase; PD-1, programmed cell death protein-1; T-VEC, talimogene laherparepvec; ULN, upper limit of normal.

^a LDH based on maximum value on or before dosing.
^b Multiple occurrences within a category were counted only once for each patient. Multiple categories of prior PD-1 therapy for a given patient were counted.

^c Percentages are based on the number of patients in the analysis set.
^d The maximum lines of therapy the patient ever received was counted; a single patient was included in one subcategory only.

^e Percentages are based on the number of prior anticancer therapies for current malignancy. This included all types of prior therapies including immunotherapy, which are listed in Table S3, Data Supplement, online only.

likely enhances the efficacy of PD-1 inhibitors in combination regimens by favorably altering the TME [33]. The MASTERKEY-115 trial (NCT04068181) assessed T-VEC plus pembrolizumab (T-VEC-pembrolizumab) in patients with advanced melanoma with confirmed progressive disease (PD) or recurrence on prior immunotherapy in a metastatic or adjuvant setting, respectively. Herein, we report results of the primary analysis, performed when all treated patients were followed up for ≥ 24 weeks for tumor assessment.

2. Methods

2.1. Study design

MASTERKEY-115 was an open-label, single-arm, multicenter, phase II trial evaluating T-VEC-pembrolizumab following melanoma progression on anti-PD-1 therapy (immediate prior line before enrollment) and enrolled patients into one of four cohorts (Table S1 and Fig. S1). Cohorts 1 and 2 comprised patients with unresectable/metastatic melanoma who had primary and acquired resistance, respectively, received anti-PD-1 in a recurrent/metastatic setting and experienced PD within 12 weeks of their last anti-PD-1 dose. Primary resistance was defined as a best overall response (BOR) of PD or stable disease (SD) on anti-PD-1 therapy prior to confirmed PD. Acquired resistance was defined as confirmed PD following a complete (CR) or partial (PR) response on anti-PD-1 therapy per investigator assessment. Cohorts 3 and 4 comprised patients with resectable disease who received only adjuvant anti-PD-1 and were disease-free for < 6 months (cohort 3) or ≥ 6 months (cohort 4) after starting the anti-PD-1 therapy before confirmed recurrence. All patients provided written informed consent; study protocols (online only) received institutional approval.

Table 2
Ratio of injected by uninjected lesions (target and non-target) per modified irRC-RECIST.

	T-VEC + pembrolizumab				All cohorts (N = 71)
	Primary resistance Cohort 1 (n = 26)	Acquired resistance Cohort 2 (n = 15)	Adjuvant < 6 months Cohort 3 (n = 15)	Adjuvant ≥ 6 months Cohort 4 (n = 15)	
Total number of injected lesions	75	25	40	32	172
Total number of uninjected lesions	134	79	50	37	300
Injected/uninjected lesion ratio	0.6	0.3	0.8	0.9	0.6
Median number of injected lesions	2	2	2	2	2
Median number of uninjected lesions	5	5	4	3	4
Total number of lesions with missing information about injected/uninjected	19	24	0	8	51

Data are the number of lesions unless specified otherwise. The full analysis set included all enrolled patients who had received ≥ 1 dose of T-VEC and pembrolizumab in combination. irRC-RECIST, Immune-related Response Evaluation Criteria in Solid Tumors; T-VEC, talimogene laherparepvec.

2.2. Patients

The study enrolled patients aged ≥ 18 years with histologically confirmed stage IIIB–IVM1d [34] unresectable or metastatic melanoma suitable for injection; measurable disease; an ECOG performance status of 0 or 1; adequate organ function; prior anti-PD-1 therapy for ≥ 2–3 consecutive cycles within an 8-week period and PD as defined by modified Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1) criteria; [35] and a negative pregnancy test. A protocol amendment permitted the inclusion of patients with stage IVM1d disease with a maximum of three previously treated cerebral metastases. Key exclusion criteria included disease progression on > 1 line of anti-PD-1 therapy.

2.3. Treatment

T-VEC and pembrolizumab were administered once every 3 weeks (Q3W). T-VEC was injected intralesionally at ≤ 4.0 mL of 10⁶ plaque-forming units (PFU)/mL (first dose), followed by ≤ 4.0 mL of 10⁸ PFU/mL after 3 weeks and Q3W thereafter. Pembrolizumab (200 mg, intravenous) was administered Q3W. T-VEC and pembrolizumab treatment continued until documented confirmed iPD per modified immune-related RECIST (irRC-RECIST) [36], treatment intolerance, or 102 weeks following the first dose, whichever occurred first. T-VEC treatment continued until the disappearance of all injectable lesions.

2.4. Endpoints

The primary endpoint was investigator-assessed ORR, defined as the incidence of a BOR of CR or PR, per modified RECIST v1.1 [35]. Key secondary endpoints included investigator-assessed iCRR and iPFS per

modified irRC-RECIST, and safety. Other secondary endpoints included ORR, BOR, durable response rate (DRR), duration of response (DOR), and disease control rate (DCR) per modified RECIST v1.1 and modified irRC-RECIST, as well as overall survival (OS). Exploratory lesion-level responses in target lesions were assessed.

2.5. Assessments

Clinical and radiographic tumor assessments were performed using modified RECIST v1.1 and irRC-RECIST and continued through to PD. Treatment decisions were made per modified irRC-RECIST. Adverse events (AEs) were graded using CTCAE version 5.0.

2.6. Statistical analysis

The planned sample size of 100 (cohorts 1/2/3/4, n = 30/40/15/15) assumed that if the primary endpoint, ORR, using modified RECIST v1.1, was ≥ 25%, the lower bound of the 95% confidence interval (CI) for an objective response would be > 15%. Efficacy was assessed in all enrolled patients who received ≥ 1 dose each of T-VEC and pembrolizumab in combination (full analysis set). The safety analysis set included all enrolled patients who received ≥ 1 dose of T-VEC or pembrolizumab.

Analyses were descriptive, with no formal hypothesis testing, and were performed separately for each cohort. Cohorts pooled together (all cohorts, cohorts 1 and 2, and cohorts 3 and 4) were also analyzed. A futility analysis was planned for cohorts 1 and 2. Proportions and the corresponding 95% exact Clopper-Pearson CIs were calculated. Time-to-event endpoints were estimated using the Kaplan-Meier method.

3. Results

3.1. Patients

Between January 2020 and February 2021, 72 patients were enrolled of whom 71 were treated (Figure 1). The patient enrollment was lower than the planned sample size as recruitment to this study was discontinued following the futility analysis of the MASTERKEY-265 phase III trial evaluating T-VEC-pembrolizumab in immunotherapy-naïve advanced melanoma [24]. At the data cutoff of August 19, 2021, the median (range) follow-up was 9.7 (1.0–17.5) months. Of the 72 enrolled patients (cohorts 1/2/3/4: n = 27/15/15/15), 26 (36.1%) continued study treatment (cohorts 1/2/3/4: T-VEC, n = 7/3/9/7; pembrolizumab, n = 7/2/10/7) and 25 (34.7%) had discontinued the study (Figure 1 and Table S2).

Baseline patient characteristics for the pooled cohort (all cohorts, N = 71) were as follows: median (range) age, 65 (33–89) years; stage IIIB–IVM1a/IVM1b–d disease, 47.9%/52.1%; BRAF-mutant tumors, 28.2%; and LDH > 1x ULN, 29.6% (Table 1). All patients had received prior anti-PD-1 therapy; 54 (76.1%) and 17 (23.9%) had received monotherapy with nivolumab or pembrolizumab, respectively, while 6 (8.5%) had received ipilimumab-nivolumab combination therapy (Table 1 and Table S3).

3.2. Exposure

In cohorts 1–4, respectively, the total number of injected lesions was 75, 25, 40, and 32; uninjected lesions was 134, 79, 50, and 37; and the ratio of injected by uninjected lesions was 0.6, 0.3, 0.8, and 0.9 (Table 2). The median (range) duration of T-VEC treatment was 18.1 (0.1–74.7) weeks overall, and 12.1 (0.1–70.1), 15.4 (6.1–58.1), 18.1 (3.1–54.1), and 51.1 (3.1–74.7) weeks in cohorts 1–4, respectively. For pembrolizumab, it was 24.0 (0.1–75.9) weeks overall, and 12.0 (0.1–75.9), 20.1 (6.1–63.1), 42.3 (7.0–60.1), and 55.0 (3.1–74.7) weeks in cohorts 1–4, respectively.

Table 3
Response to therapy.

Response	T-VEC + pembrolizumab				
	Primary resistance	Acquired resistance	Adjuvant < 6 months	Adjuvant ≥ 6 months	All cohorts
	Cohort 1 (n = 26)	Cohort 2 (n = 15)	Cohort 3 (n = 15)	Cohort 4 (n = 15)	(N = 71)
Response assessment per modified RECIST version 1.1					
Objective response rate (CR/PR)	0	1 (6.7)	6 (40.0)	7 (46.7)	14 (19.7)
95% CI	NE	(0.2–32.0)	(16.3–67.7)	(21.3–73.4)	(11.2–30.9)
Best overall response					
Complete response	0	0	2 (13.3)	3 (20.0)	5 (7.0)
Partial response	0	1 (6.7)	4 (26.7)	4 (26.7)	9 (12.7)
Stable disease	8 (30.8)	4 (26.7)	0	6 (40.0)	18 (25.4)
Progressive disease	11 (42.3)	5 (33.3)	9 (60.0)	1 (6.7)	26 (36.6)
Unevaluable	0	1 (6.7)	0	0	1 (1.4)
Non-CR/Non-PD	1 (3.8)	0	0	0	1 (1.4)
Not done	6 (23.1)	4 (26.7)	0	1 (6.7)	11 (15.5)
Complete response rate	0	0	2 (13.3)	3 (20.0)	5 (7.0)
95% CI	NE	NE	(1.7–40.5)	(4.3–48.1)	(2.3–15.7)
Disease control rate (CR/PR/SD)	8 (30.8)	5 (33.3)	6 (40.0)	13 (86.7)	32 (45.1)
95% CI	(14.3–51.8)	(11.8–61.6)	(16.3–67.7)	(59.5–98.3)	(33.2–57.3)
Durable response rate	0	0	2 (13.3)	4 (26.7)	6 (8.5)
95% CI	NE	NE	(1.7–40.5)	(7.8–55.1)	(3.2–17.5)
PFS					
Deaths or disease progression, number of events (%)	17 (65.4)	11 (73.3)	9 (60.0)	9 (60.0)	46 (64.8)
Median ^f , months	3.2	4.8	2.8	13.8	4.9
95% CI	(2.8– 5.5)	(2.4– 13.9)	(2.7–NE)	(5.6–16.5)	(2.8–10.9)
PFS at 12 months ^f , %	18.2	32.0	40.0	55.0	34.0
95% CI	(4.2–39.9)	(10.9–55.7)	(16.5–62.8)	(24.4–77.6)	(21.8–46.5)
Response assessment per modified irRC-RECIST					
Objective response rate (iCR/iPR)	1 (3.8)	1 (6.7)	8 (53.3)	7 (46.7)	17 (23.9)
95% CI	(0.1–19.6)	(0.2–32.0)	(26.6–78.7)	(21.3–73.4)	(14.6–35.5)
Best overall response					
Complete response	0	0	2 (13.3)	2 (13.3)	4 (5.6)
Partial response	1 (3.8)	1 (6.7)	6 (40.0)	5 (33.3)	13 (18.3)
Stable disease	12 (46.2)	5 (33.3)	3 (20.0)	6 (40.0)	26 (36.6)
Progressive disease	5 (19.2)	4 (26.7)	1 (6.7)	0	10 (14.1)
Unevaluable	2 (7.7)	1 (6.7)	2 (13.3)	1 (6.7)	6 (8.5)
Not done	6 (23.1)	4 (26.7)	1 (6.7)	1 (6.7)	12 (16.9)
Complete response rate	0	0	2 (13.3)	2 (13.3)	4 (5.6)
95% CI	NE	NE	(1.7–40.5)	(1.7–40.5)	(1.6–13.8)
Disease control rate (iCR/iPR/iSD)	13 (50.0)	6 (40.0)	11 (73.3)	13 (86.7)	43 (60.6)
95% CI	(29.9–70.1)	(16.3–67.7)	(44.9–92.2)	(59.5–98.3)	(48.2–72.0)
Durable response rate	1 (3.8)	0	2 (13.3)	4 (26.7)	7 (9.9)
95% CI	(0.1–19.6)	NE	(1.7–40.5)	(7.8–55.1)	(4.1–19.3)
Progression-free survival					
Deaths or disease progression, number of events (%)	11 (42.3)	10 (66.7)	4 (26.7)	2 (13.3)	27 (38.0)
Median ^f , months	5.5	8.2	NE	NE	15.0
95% CI	(2.8–NE)	(2.7–15.0)	(2.7–NE)	NE	(8.2–NE)
PFS at 12 months ^f , %	46.2	44.4	66.7	85.7	59.9
95% CI	(24.1–65.7)	(18.5–67.7)	(30.3–87.1)	(53.9–96.2)	(46.3–71.1)

Data are the number (%) of patients unless specified otherwise. The full analysis set included all enrolled patients who had received ≥ 1 dose of T-VEC and pembrolizumab in combination. All CIs reported are binomial proportion with exact 95% CIs. iCR, iPR, and iSD denote CR, PR, and SD per irRC-RECIST, respectively. CI, confidence interval; CR, complete response; irRC-RECIST, immune-related RECIST; NE, not estimable; PD, progressive disease; PFS, progression-free survival; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors; SD, stable disease; T-VEC, talimogene laherparepvec.

^f Estimated using the Kaplan-Meier method.

3.3. Efficacy

The pooled ORR (95% CI) per RECIST v1.1 was 19.7% (11.2–30.9), with 7.0% and 12.7% of patients achieving CR and PR, respectively. The ORR (95% CI) in cohorts 1–4 respectively, was 0%, 6.7% (0.2–32.0), 40.0% (16.3–67.7), and 46.7% (21.3–73.4), with CR in 0%, 0%, 13.3%, and 20.0% of patients (Table 3). When assessed per irRC-RECIST, the pooled iORR (95% CI) was 23.9% (14.6–35.5), with 5.6% of patients achieving iCR and 18.3% achieving iPR. The iORR (95% CI) in cohorts 1–4, respectively, was 3.8% (0.1–19.6), 6.7% (0.2–32.0), 53.3% (26.6–78.7), and 46.7% (21.3–73.4), with iCR in 0%, 0%, 13.3%, and 13.3% of patients (Table 3). The duration of prior treatment and cohort-specific BORs are presented in Fig. S2. Lesion-level BORs are reported in Table S4.

The iDCR (95% CI) was 50.0% (29.9–70.1), 40.0% (16.3–67.7), 73.3% (44.9–92.2), and 86.7% (59.5–98.3) in cohorts 1–4, respectively

(Table 3). Median iDOR was not reached in any cohort (pooled cohort, range: 2.73 + to 11.07 +, where “+” indicates the value is a censoring time); 17 (23.9%) patients with a response remained in response, with 1 (3.8%), 1 (6.7%), 8 (53.3%), and 7 (46.7%) responders in cohorts 1–4, respectively.

Median (95% CI) iPFS was 15.0 months (8.2–not estimable [NE]) overall, with an estimated 12-month iPFS of 59.9% (95% CI, 46.3–71.1) (Table 3). Median (95% CI) iPFS was 5.5 (2.8–NE) and 8.2 (2.7–15.0) months in cohorts 1 and 2, respectively, and was not reached in cohorts 3 and 4 (Figure 2).

Median OS was not reached, with 49 of 71 (69.0%) patients alive at the data cutoff.

3.4. Safety

Any-grade treatment-related AEs (TRAEs) were reported in 54

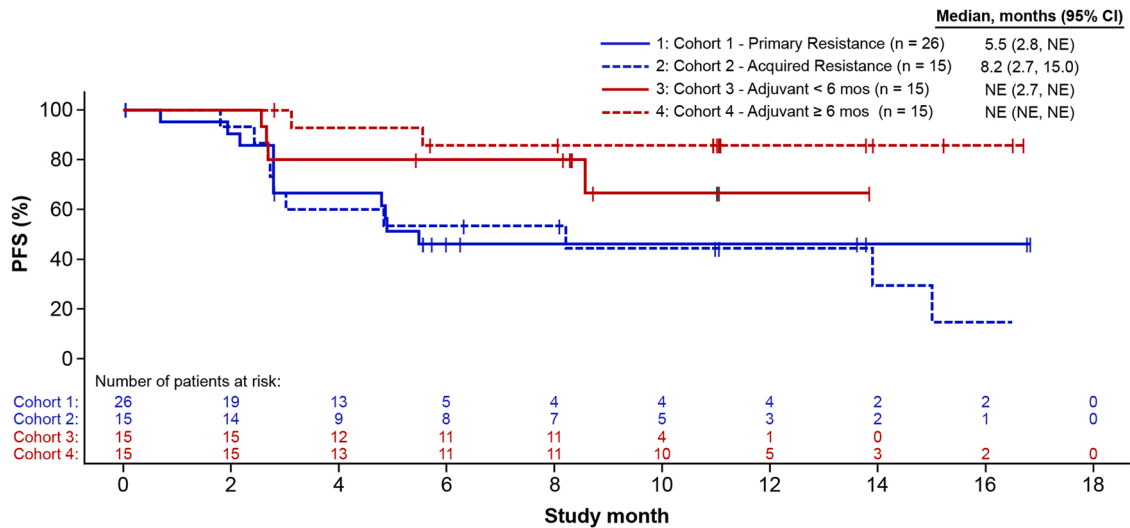


Fig. 2. Kaplan-Meier estimate of progression-free survival per modified irRC-RECIST in all cohorts combined. The full analysis set included all enrolled patients who had received ≥ 1 dose of T-VEC and pembrolizumab in combination. iPFS per modified irRC-RECIST was defined as the interval from the first dose to the earlier event of iPD or death from any cause. Patients without an event were censored at their last evaluable post-baseline tumor assessment. Vertical lines indicate censoring. CI, confidence interval; iPD, progressive disease by modified irRC-RECIST; iPFS, progression-free survival by modified irRC-RECIST; irRC-RECIST, immune-related RECIST; RECIST, Response Evaluation Criteria in Solid Tumors; NE, not estimable; T-VEC, talimogene laherparepvec.

(76.1%) patients overall, and in 17 (65.4%), 10 (66.7%), 14 (93.3%), and 13 (86.7%) patients in cohorts 1–4, respectively; the most common ($\geq 15\%$ patient incidence in the pooled cohort) were pyrexia (29.6%), fatigue, and influenza-like illness (15.5% each) (Table 4). Grade ≥ 3 TRAEs and fatal AEs occurred in 9 (12.7%) and 10 (14.1%) patients, respectively (Table S5). One (6.7%) death reported in cohort 4, due to natural causes, was attributed possibly to T-VEC treatment by the investigator.

4. Discussion

In MASTERKEY-115, T-VEC-pembrolizumab showed antitumor activity in selected patients with advanced melanoma that progressed on prior anti-PD-1 agents. T-VEC-pembrolizumab yielded ORRs of 40.0–46.7% in patients with PD in the adjuvant setting (cohorts 3/4) and 0–6.7% in the metastatic setting (cohorts 1/2), with manageable toxicity across all cohorts. These data are in accordance with those reported recently from the IGNYTE study that evaluated RP1 (an HSV-1-based oncolytic immunotherapy) plus nivolumab in a patient population similar to this study. Higher ORRs were observed in patients with PD-1–refractory melanoma who had received prior anti-PD-1 in the adjuvant setting than those in the non-adjuvant setting (50.0% versus 31.1%) [37].

A potential reason for the promising responses observed in cohorts 3/4 and the lack thereof in cohorts 1/2 may be related to the ability of T-VEC to mediate superior direct (local tumor lysis following direct intratumoral injection) than indirect (distant systemic immune-mediated) antitumor effect [38]. This could result from the heterogeneity of immunogenic mutations between metastases. Indeed, in the post-surgical adjuvant setting, lesions injected with T-VEC are more likely to be representative of the patients' overall mutation burden than in patients with widespread metastases, which can have highly disparate mutation burden. As such, T-VEC injection may prime immunity against the injected lesions' specific immunogenic mutation burden and not against the distinct mutation burden of more distant metastases. Indeed, the ratios of patients with skin and lymph node involvement only (stages IIIB/IVM1a) were higher in cohorts 3 and 4 (60.0% and 66.7%) than in cohorts 1 and 2 (30.8% and 46.7%). Furthermore, this may also be the reason for the lack of substantial abscopal responses historically seen with radiation, or the reason why TIL therapy has only a 31.4% ORR in

PD-1–refractory stage IV melanoma, as the resected tumor used for therapy might not elicit a robust antitumor immune response for a patient's overall cancer burden [39,40]. The ratios of injected by uninjected lesions were numerically higher in cohorts 3/4 than in cohorts 1/2, supporting that a relatively lower tumor burden, with the possibility to directly target a large ratio of metastases, may yield improved outcomes with T-VEC treatment as observed in cohorts 3/4.

Our study demonstrated that T-VEC-pembrolizumab yielded promising iDCRs (40.0–86.7%), suggesting that T-VEC may elicit a meaningful long-term systemic anticancer effect. This is supported by findings of the largest randomized neoadjuvant trial for an oncolytic virus in advanced melanoma that showed durable improvements in 5-year survival rates with neoadjuvant T-VEC plus surgery over standard surgery (recurrence-free survival, 22.3% for neoadjuvant T-VEC and 15.2% for surgery [overall HR: 0.76; 80% CI: 0.60, 0.97]).

The ORR/iORR observed with T-VEC-pembrolizumab in the metastatic cohorts 1/2 was 0%/3.8% in the primary resistance setting and 6.7%/6.7% in the acquired resistance setting. In the LEAP-004 study, the ORR for lenvatinib-pembrolizumab was generally similar across settings (primary metastatic, 22.6%; secondary metastatic, 22.7%) [15]. As such, the ORRs with lenvatinib-pembrolizumab are higher than those with T-VEC-pembrolizumab, the reasons for which could be related to the different mechanisms of action of the PD-1 combination partner, dissimilar study designs and patient populations, and different prior lines of therapy [15]. Collectively, T-VEC-pembrolizumab likely elicits different response rates across clinical resistance phenotypes. These data require cautious interpretation given the small cohort sizes and because the study was not powered to compare between cohorts.

The median iDOR was not reached in any cohort, with more responders in cohorts 3/4 than in cohorts 1/2. The median PFS in this study for the pooled cohort (4.9 months) was generally similar to that reported in the ICI-refractory setting for pembrolizumab-ipilimumab (5 months; 95% CI, 2.8–8.3), lenvatinib-pembrolizumab (4.2 months; 95% CI, 3.8–7.1), and longer than that reported for nivolumab-relatlimab (2.1 [95% CI, 1.9–3.5] to 3.2 months [95% CI, 1.9–3.6]), although such cross-trial comparisons are constrained by the caveats described earlier [12,15,17,24]. The median (95% CI) PFS for pooled cohorts 1 and 2 in this study was 3.6 months (2.8–5.5; data not shown), which is comparable to that reported for lenvatinib-pembrolizumab [15]. The iDCR for pooled cohorts 1 and 2 was 46.3% (95% CI, 30.7–62.6; data not

Table 4
Treatment-emergent and treatment-related adverse events.

Adverse events	T-VEC + pembrolizumab									
	Primary resistance		Acquired resistance		Adjuvant < 6 months		Adjuvant ≥ 6 months		All cohorts	
	Cohort 1 (n = 26)		Cohort 2 (n = 15)		Cohort 3 (n = 15)		Cohort 4 (n = 15)		(N = 71)	
Any-grade TEAEs	24 (92.3)		15 (100.0)		15 (100.0)		14 (93.3)		68 (95.8)	
Grade ≥ 3 TEAEs	9 (34.6)		7 (46.7)		3 (20.0)		7 (46.7)		26 (36.6)	
Serious TEAEs	10 (38.5)		6 (40.0)		3 (20.0)		5 (33.3)		24 (33.8)	
Fatal TEAEs	4 (15.4)		3 (20.0)		1 (6.7)		2 (13.3)		10 (14.1)	
Any-grade TRAEs	17 (65.4)		10 (66.7)		14 (93.3)		13 (86.7)		54 (76.1)	
Grade ≥ 3 TRAEs	1 (3.8)		2 (13.3)		0		6 (40.0)		9 (12.7)	
Fatal TRAEs	0		0		0		1 (6.7) [§]		1 (1.4)	
TRAEs	Any grade	Grade ≥ 3	Any grade	Grade ≥ 3	Any grade	Grade ≥ 3	Any grade	Grade ≥ 3	Any grade	Grade ≥ 3
Pyrexia	7 (26.9)		4 (26.7)		5 (33.3)		5 (33.3)		21 (29.6)	
Chills	5 (19.2)		2 (13.3)				2 (13.3)		10 (14.1)	
Asthenia	3 (11.5)							1 (6.7)		1 (1.4)
Fatigue	3 (11.5)		2 (13.3)		5 (33.3)			1 (6.7)	11 (15.5)	1 (1.4)
Injection site pain	3 (11.5)		2 (13.3)		4 (26.7)				9 (12.7)	
Nausea	3 (11.5)		3 (20.0)		2 (13.3)		2 (13.3)		10 (14.1)	
Vomiting	3 (11.5)						2 (13.3)			
Vitiligo			2 (13.3)				2 (13.3)			
Influenza like illness					6 (40.0)		2 (13.3)		11 (15.5)	
Headache					4 (26.7)					
Hyperthermia					2 (13.3)					
Injection site reaction					2 (13.3)					
Lethargy					2 (13.3)					
Myalgia					2 (13.3)					
Pruritus					2 (13.3)					
Diarrhea							5 (33.3)	2 (13.3)	9 (12.7)	2 (2.8)
Arthralgia							3 (20.0)	1 (6.7)		1 (1.4)
ALT increased		1 (3.8)					2 (13.3)	1 (6.7)		2 (2.8)
Anemia							2 (13.3)	1 (6.7)		1 (1.4)
Colitis							2 (13.3)	1 (6.7)		1 (1.4)
Eczema							2 (13.3)			
Hepatic cytolysis				1 (6.7)						1 (1.4)
Lipase increased				1 (6.7)						1 (1.4)
Death								1 (6.7)		1 (1.4)
Diarrhea infectious								1 (6.7)		1 (1.4)
Pancreatitis								1 (6.7)		1 (1.4)
Septic shock								1 (6.7)		1 (1.4)

Data are the number (%) of patients. The safety analysis set included all enrolled patients who received ≥ 1 dose each of T-VEC or pembrolizumab. TEAEs were defined as any AE occurring after initiation of the first dose of study treatment through 30 days after the last administration of study treatment. Serious AEs were defined as any AE observed by the investigator or reported by the patient that occurred after 90 (+ 7) days after the cessation of all study treatment or 30 (+ 7) days following cessation of treatment if the patient initiated new anticancer therapy, whichever was earlier. AEs were coded using MedDRA version 24.0. The CTCAE version 5.0 was used to grade severity of AEs. TRAEs are those TEAEs deemed related to study treatment as per investigator as reported on the eCRF. Fatal TEAEs are all TEAEs that were fatal and occurred between first dose day and up to 30 days after the last dose of study drug. Any-grade TRAEs (related to any study treatment) listed above occurred in ≥ 10% of patients receiving T-VEC-pembrolizumab in any cohort. All grade ≥ 3 TRAEs for T-VEC-pembrolizumab are listed. AE, adverse event; ALT, alanine aminotransferase; CTCAE, Common Terminology Criteria for Adverse Events; eCRF, electronic case report form; MedDRA, Medical Dictionary for Regulatory Activities; TEAE, treatment-emergent AE; TRAE, treatment-related AE; T-VEC, talimogene laherparepvec.

[§] One fatal TRAE reported in cohort 4 was due to natural causes which was attributed to T-VEC by the investigator.

shown). Collectively, these data suggest a potential benefit with T-VEC-pembrolizumab in the metastatic setting as well.

The safety profile of T-VEC-pembrolizumab was consistent with the defined AE spectrum for either agent alone or the combination in the treatment-naïve settings [21,24,41].

Our study has the intrinsic limitations of a single-arm, phase II trial, had relatively small cohort sizes and a modest median follow-up duration of 9.7 months, and was not designed to formally compare cohorts. Furthermore, we cannot distinguish whether the effectiveness of the combination could be attributed to T-VEC alone or to the addition of pembrolizumab to T-VEC, given the discordance between efficacy findings from the phase III T-VEC monotherapy and combination therapy studies in PD-1-naïve melanoma. While T-VEC monotherapy yielded higher response rates versus GM-CSF in OPTiM, T-VEC-pembrolizumab did not show statistically superior efficacy versus placebo plus pembrolizumab in MASTERKEY-265 [21,24]. Given the similarities between the patient populations for MASTERKEY-265 and MASTERKEY-115, it was the sponsor decision to halt new screening into MASTERKEY-115 following the knowledge that the futility boundary for OS was crossed for MASTERKEY-265 study despite the

acceptable safety profile for this combination in patients with unresectable stage IIIB-IVM1c melanoma. This study included a high proportion of patients with stage IVM1b–d disease, as well as a substantial percentage of those with elevated LDH, which is representative of patients with stage IV melanoma in most clinical trials [12,15,24]. The ability of T-VEC-pembrolizumab to benefit patients who experience relapse on adjuvant anti-PD-1 would be highly valuable as most patients with stage IIB–C and stage III melanoma, who are not candidates for multiple surgeries but who have a high ratio of injectable metastases, receiving adjuvant anti-PD-1 frequently develop disease recurrence [42]. With the FDA approval of adjuvant pembrolizumab and nivolumab, the number of patients reporting adjuvant therapy failure will likely increase [9,10,42]. An efficacious treatment option with limited toxicities for these patients with PD-1-resistant melanoma is critical.

In conclusion, T-VEC-pembrolizumab demonstrated antitumor activity and tolerability in patients with PD-1-refractory advanced melanoma who were previously treated in the adjuvant setting, when a high ratio of lesions can be injected, than in the metastatic setting. These data support the potential of T-VEC-pembrolizumab to help overcome immunotherapy resistance in patients who experience disease

progression or recurrence on current SOC.

Ethics Approval and Consent to Participate

Study procedures were approved by the institutional review boards or ethics committees of each participating site, and all patients provided written informed consent.

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CRedit authorship contribution statement

Caroline Robert: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Brian Gastman:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Helen Gogas:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Piotr Rutkowski:** Investigation, Methodology, Writing – original draft, Writing – review & editing. **Georgina V. Long:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Marya F. Chaney:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Harshada Joshi:** Formal analysis, Writing – original draft, Writing – review & editing. **Yu-Lin Lin:** Formal analysis, Writing – original draft, Writing – review & editing. **Wendy Snyder:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Jason Alan Chesney:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. .

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: **Caroline Robert:** Consulting or advisory role: Bristol-Myers Squibb (BMS), Roche, Novartis, Pierre Fabre, MSD, Sanofi, AstraZeneca, Pfizer, Sunpharma; Scientific co-founder of Ribonex. **Brian Gastman:** Consulting or advisory role: Quest Imaging, Castle Biosciences; Speakers' Bureau: Castle Biosciences; Stock and other ownership interests: Castle Biosciences; Research funding: Alkermes, Neo-ImmuneTech, Merck, Instil Bio. **Helen Gogas:** Advisory board: BMS, MSD, Pierre Fabre, Sanofi; Invited Speaker: BMS, MSD, Novartis, Pierre Fabre, Sanofi; Local PI: Amgen, Array Biopharma, AstraZeneca, Bayer, BioNTech, Boehringer Ingelheim, BMS, Evaxion, Hexal AG (Sandoz Company), Highlight Therapeutics S.L., IOBiotech, Immunocore, Inovvent Biologics USA, Merck Sharpe & Dohme, Novartis, PHMR Ltd, Pierre Fabre, Regeneron, Scancell; Payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events: BMS, Pierre Fabre. **Marya F. Chaney:** Employee: Employee of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA; Stock Ownership: Merck & Co., Inc., Rahway, NJ, USA. **Harshada Joshi:** Employee: Parexel. **Yu-Lin Lin:** Employee/stock ownership: Amgen. **Wendy Snyder:** Employee/stock ownership: Amgen. **Jason A.**

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Data Sharing Statement

Qualified researchers may request data from Amgen clinical studies. Complete details are available at the following: <http://www.amgen.com/datasharing>.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ejca.2024.114120](https://doi.org/10.1016/j.ejca.2024.114120).

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